

Analyse et Prédiction des Maladies Cardiaques

Statistiques des modèles - Ensemble d'entraînement

Version corrigée - Avril 2025

Performances des modèles - Ensemble d'entraînement

Modèle	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	82.1%	84.2%	79.0%	81.5%	91.2%
Decision Tree	85.2%	90.1%	79.0%	84.2%	89.5%
KNN	84.6%	88.9%	79.0%	83.7%	91.8%
Random Forest	84.6%	86.8%	81.5%	84.1%	95.2%
AdaBoost	75.9%	76.9%	74.1%	75.5%	86.8%
SVM	87.7%	90.7%	84.0%	87.2%	94.4%
Voting Ensemble	87.7%	90.7%	84.0%	87.2%	94.1%

Détails - Logistic Regression

Métriques principales

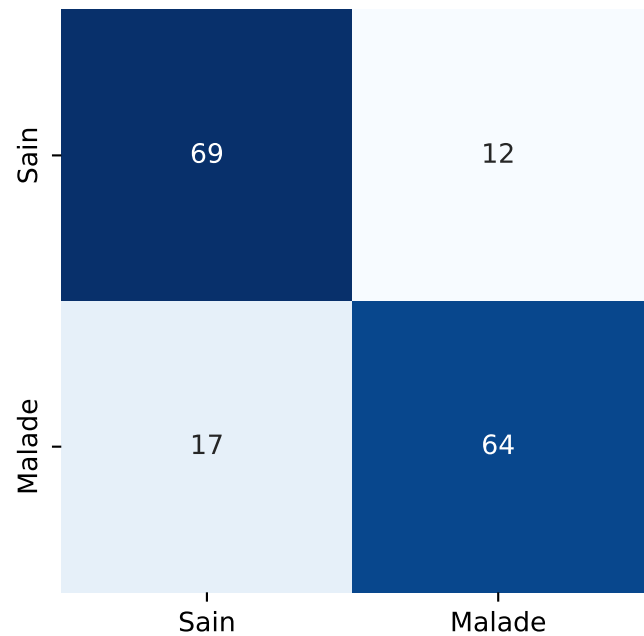
Accuracy: 82.1%

Precision: 84.2%

Recall: 79.0%

F1-Score: 81.5%

Matrice de confusion



Détail matrice

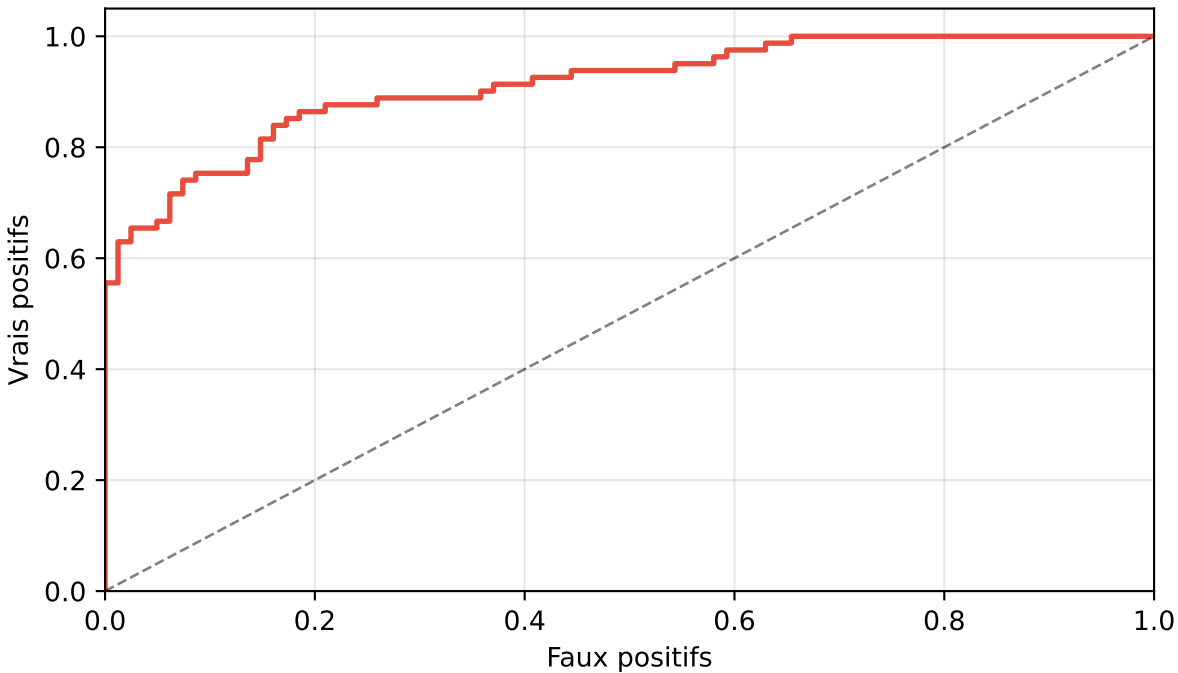
TP: 64

FP: 12

TN: 69

FN: 17

ROC (AUC = 0.912)



Détails - Decision Tree

Métriques principales

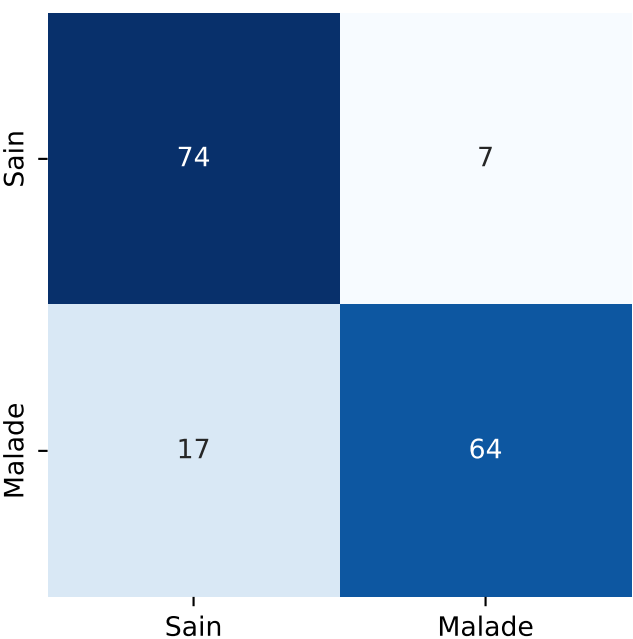
Accuracy: 85.2%

Precision: 90.1%

Recall: 79.0%

F1-Score: 84.2%

Matrice de confusion



Détail matrice

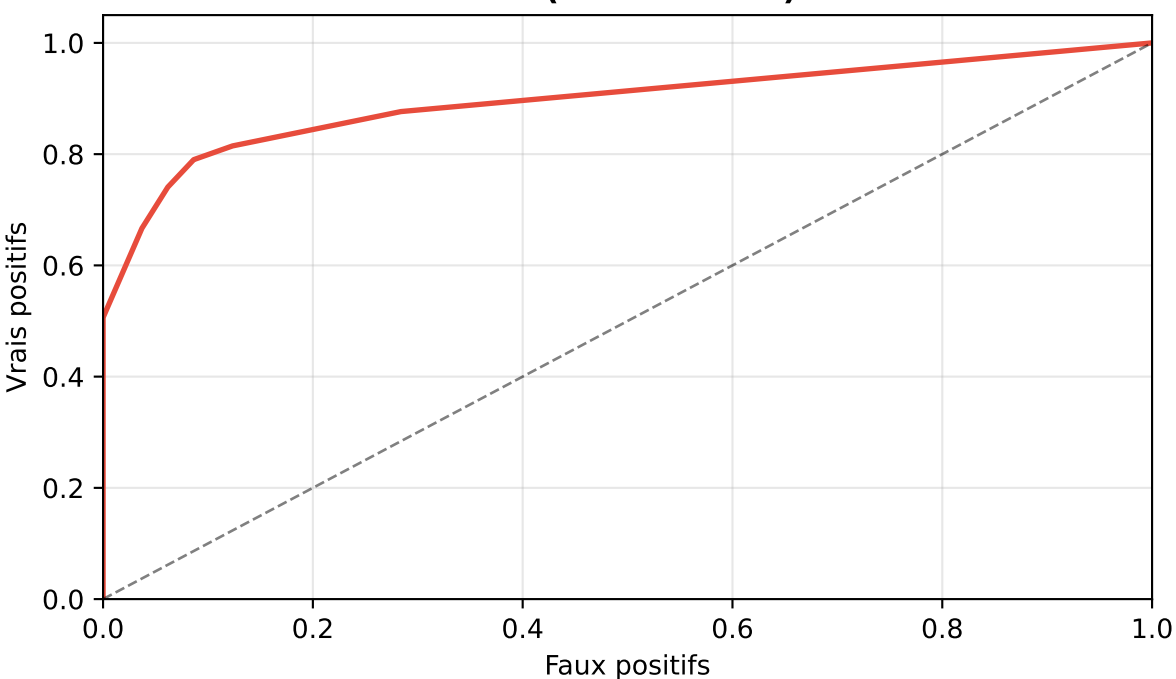
TP: 64

TN: 74

FP: 7

FN: 17

ROC (AUC = 0.895)



Détails - KNN

Métriques principales

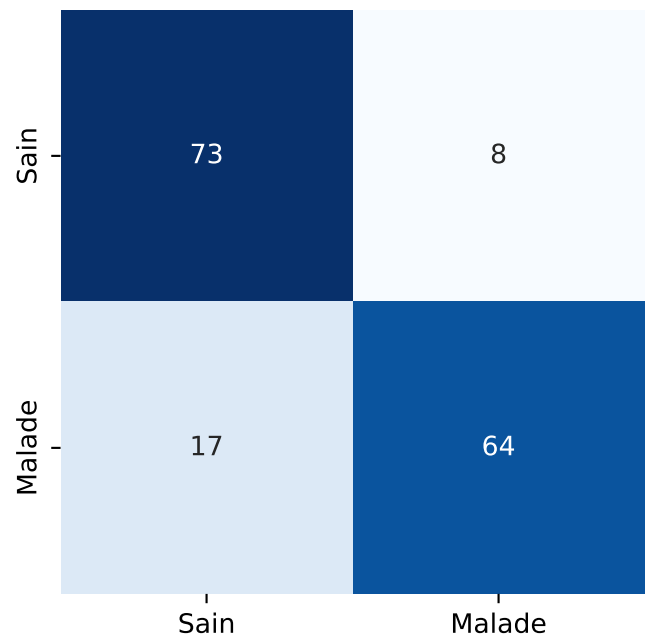
Accuracy: 84.6%

Precision: 88.9%

Recall: 79.0%

F1-Score: 83.7%

Matrice de confusion



Détail matrice

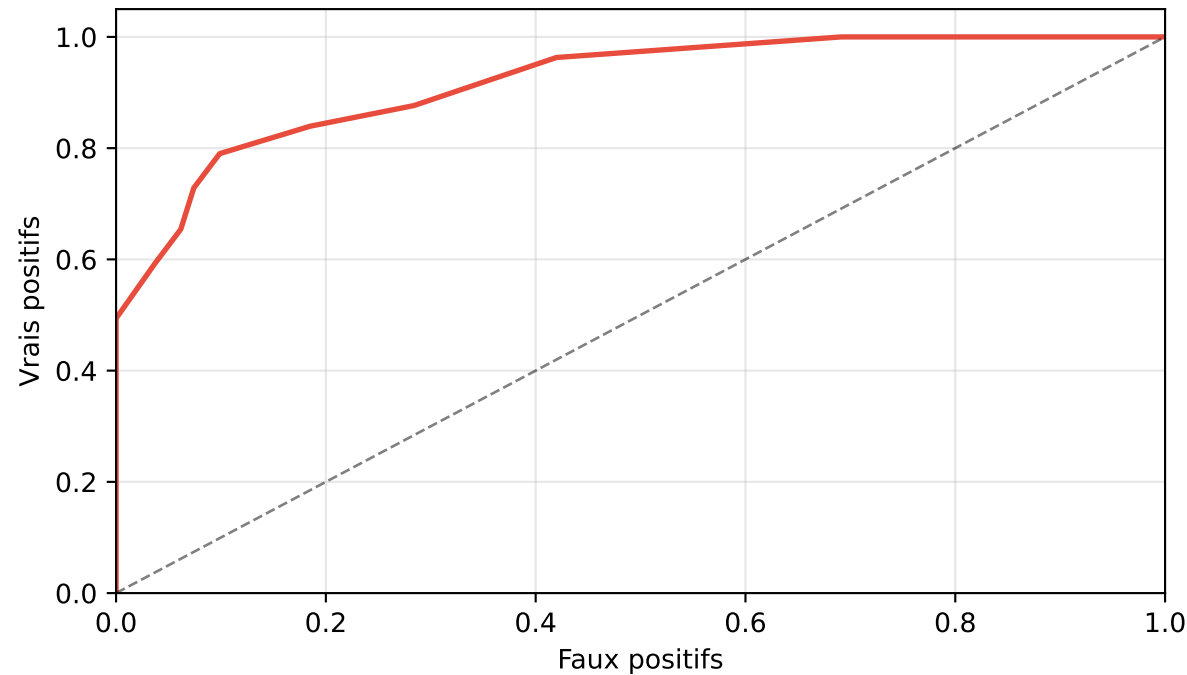
TP: 64

FP: 8

TN: 73

FN: 17

ROC (AUC = 0.918)



Détails - Random Forest

Métriques principales

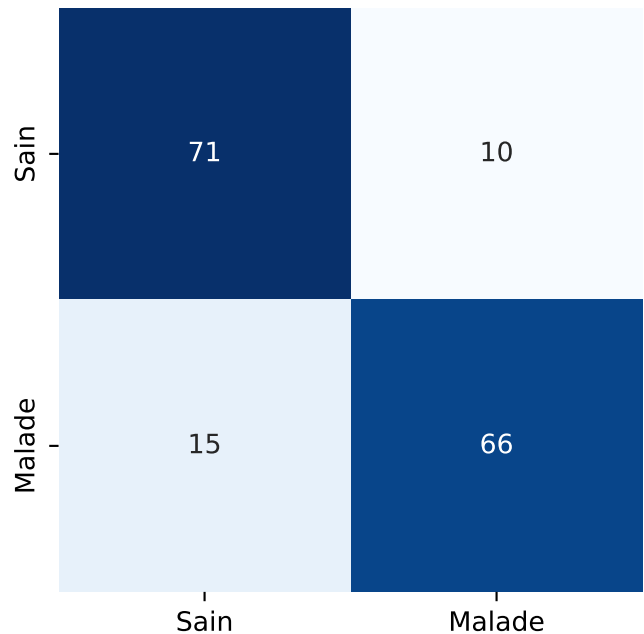
Accuracy: 84.6%

Precision: 86.8%

Recall: 81.5%

F1-Score: 84.1%

Matrice de confusion



Détail matrice

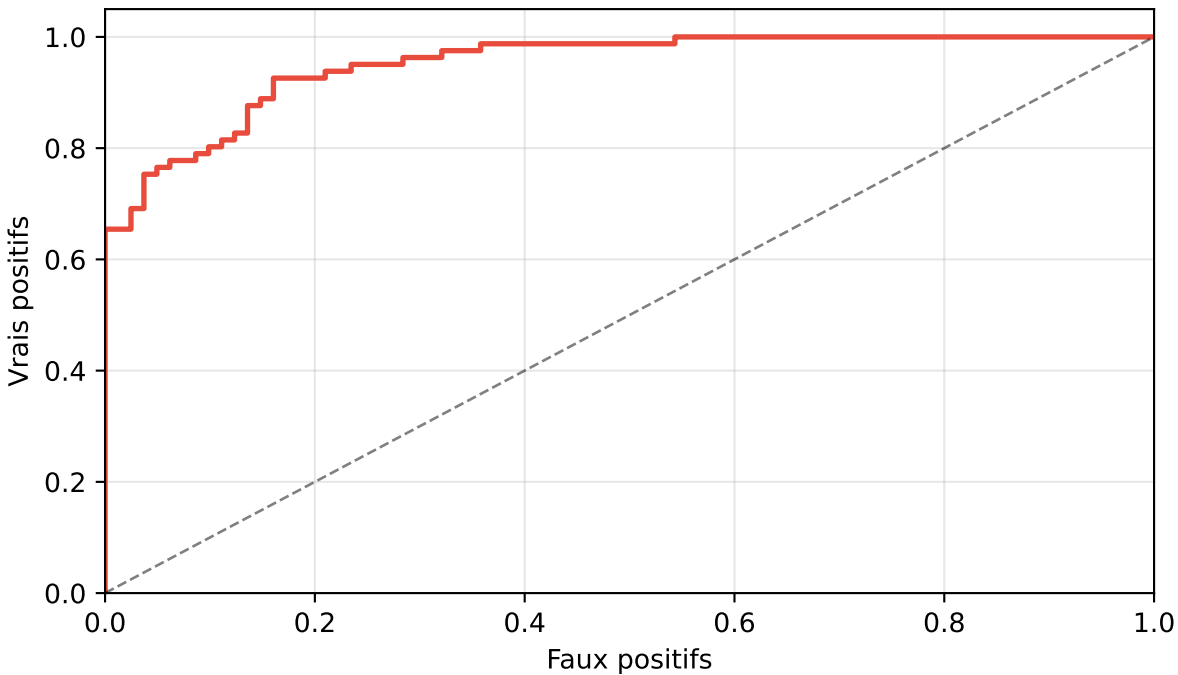
TP: 66

TN: 71

FP: 10

FN: 15

ROC (AUC = 0.952)



Détails - AdaBoost

Métriques principales

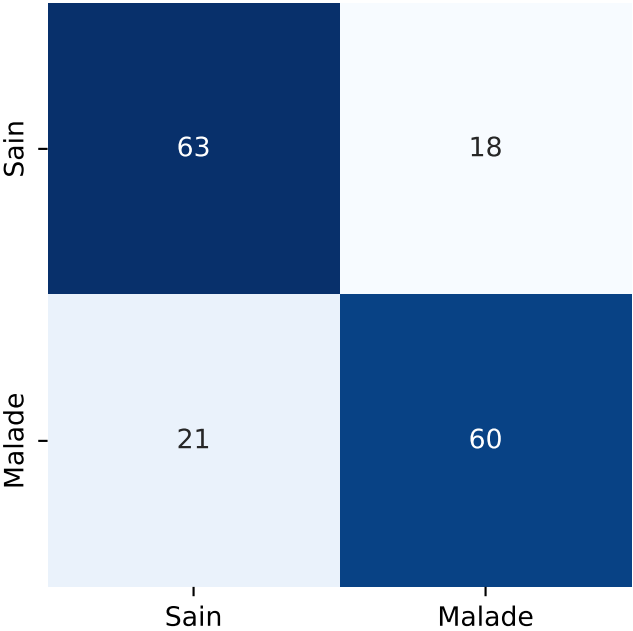
Accuracy: 75.9%

Precision: 76.9%

Recall: 74.1%

F1-Score: 75.5%

Matrice de confusion



Détail matrice

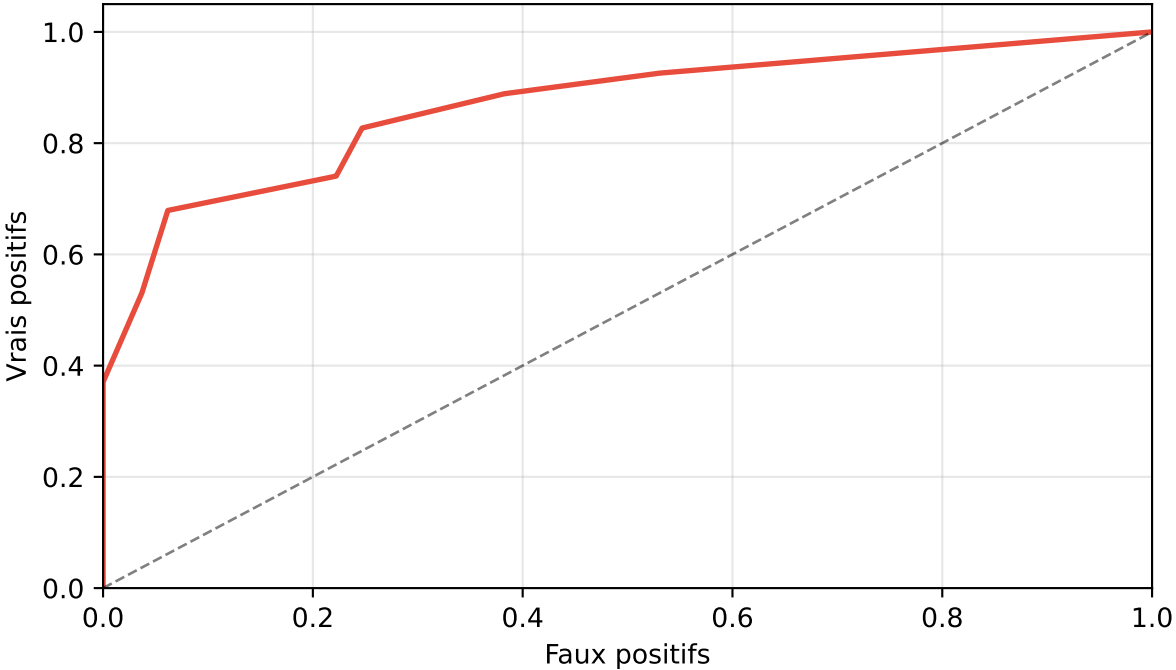
TP: 60

FP: 18

TN: 63

FN: 21

ROC (AUC = 0.868)



Détails - SVM

Métriques principales

Accuracy: 87.7%

Precision: 90.7%

Recall: 84.0%

F1-Score: 87.2%

Détail matrice

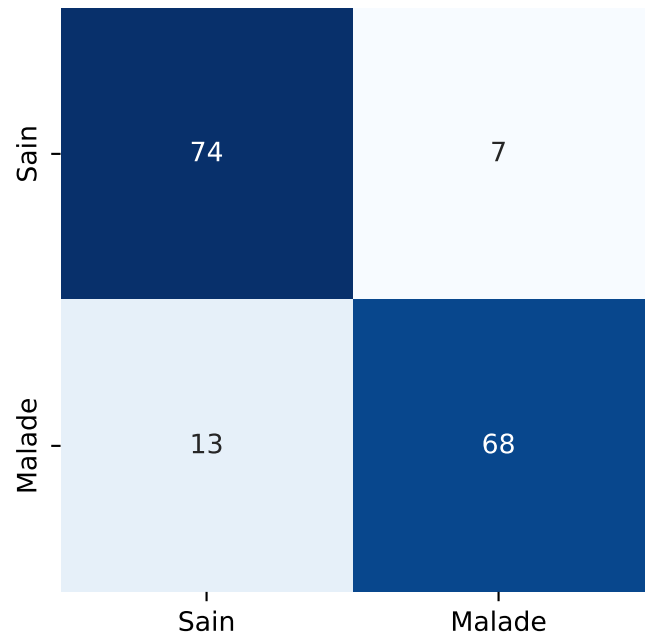
TP: 68

FP: 7

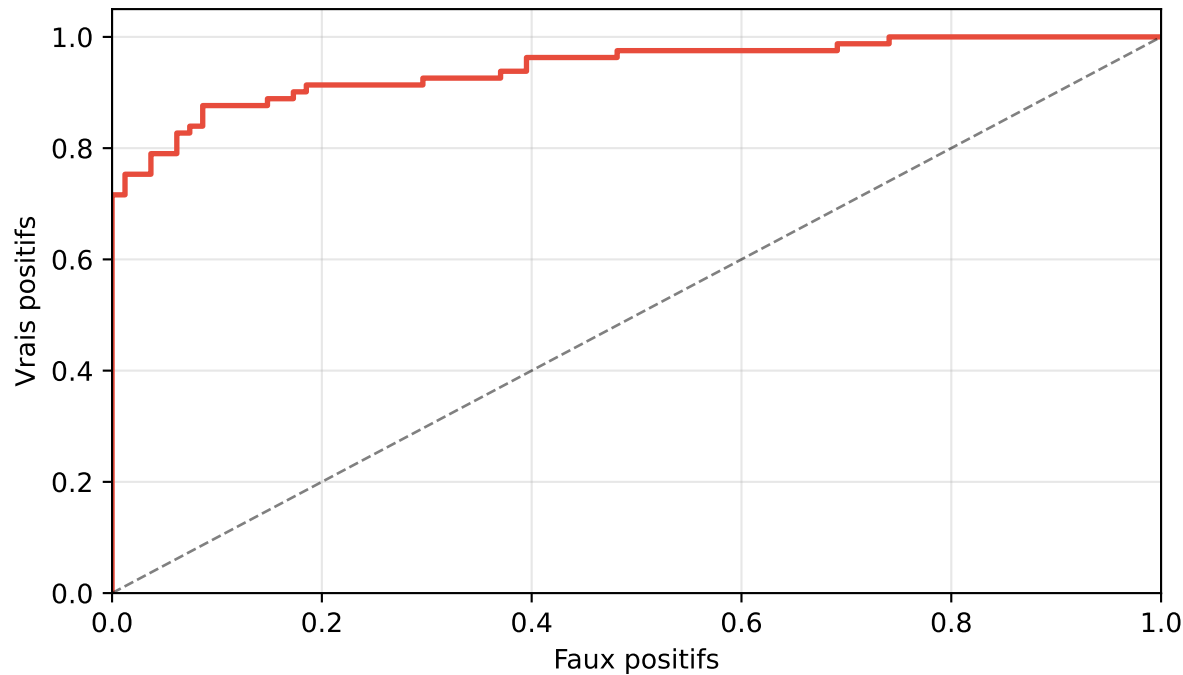
TN: 74

FN: 13

Matrice de confusion



ROC (AUC = 0.944)



Détails - Voting Ensemble

Métriques principales

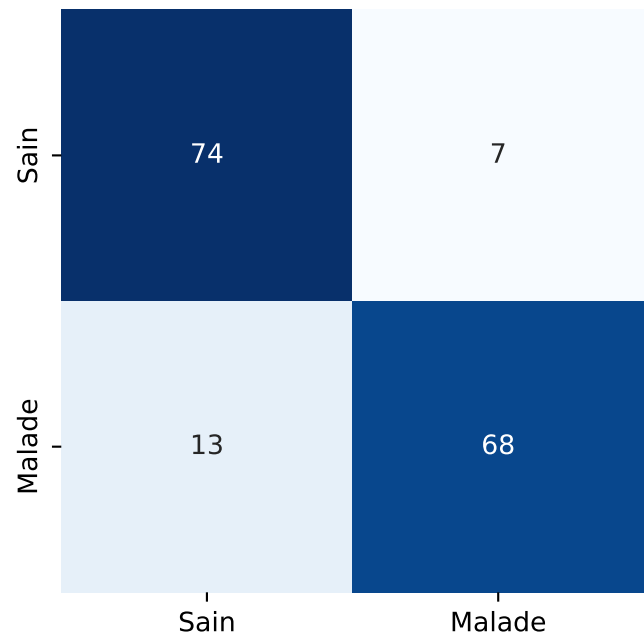
Accuracy: 87.7%

Precision: 90.7%

Recall: 84.0%

F1-Score: 87.2%

Matrice de confusion



Détail matrice

TP: 68

FP: 7

TN: 74

FN: 13

ROC (AUC = 0.941)

